When Does Erasure Coding Beat Retransmission for RDMA?

Decode Cost Decides at Microsecond RTTs

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Abstract

Forward erasure coding (FEC) can hide packet loss from RDMA transfers without waiting for a retransmission. Recent work shows it beats selective repeat over long-haul links with millisecond round-trip times (RTTs), but it does not account for the cost of decoding. Inside a datacenter, RTTs are a few microseconds, so decoding time competes directly with recovery by retransmission. We measure this cost and its effect. Intel ISA-L, a widely deployed Reed–Solomon library, inverts a $k \times k$ matrix on every new loss pattern. On a Sapphire Rapids server this takes 21 µs at k=16 and 160 µs at k=32. Erasure-count decoding, a known technique that inverts only an $m' \times m'$ matrix for the m' lost packets, takes 0.8–3.4 µs for up to four lost packets on the same SIMD kernel. Using a validated discrete-event simulator, we show that this difference decides the outcome. For a 64-flow collective step at 10^{-3} loss, FEC with ISA-L's decoder loses to selective repeat with a tail-loss probe at every RTT up to 10 µs. With erasure-count decoding it wins at every RTT, by 1.2–3.0×. The same reversal holds under bursty loss calibrated to production measurements. The benchmark and simulator are open source.

1 Introduction

Large-scale training synchronises gradients with collective operations over RDMA. One step of a collective finishes only when its slowest flow finishes, so a single lost packet can stall the whole step. RoCE NICs recover losses by retransmission: go-back-N on commodity hardware [4, 8], or selective repeat in proposals such as IRN [8]. Both need at least one RTT, and a lost *final* packet triggers no out-of-order arrival, so it waits for a timeout.

Forward erasure coding avoids the round trip by sending parity packets. SDR-RDMA [6] shows that erasure coding beats selective repeat for RDMA over long-haul links, where the RTT is 25 ms. That analysis assumes encoding overlaps transmission and does not measure decoding. Over 25 ms RTTs that is reasonable; inside a datacenter it is not. With RTTs of a few microseconds, a decoder that takes tens of microseconds can be slower than simply retransmitting.

This paper asks a narrower question than prior work: *at intra-datacenter RTTs, does the choice of Reed–Solomon decoder decide whether FEC beats retransmission?* We find that it does.

- We measure decode latency for ISA-L [5], cm256 [11] and erasure-count decoding [7, 10] on ISA-L's kernel, on a server and a laptop (§4). ISA-L's cold decode costs 21–160 µs on the server, against 0.8–3.4 µs for erasure-count decoding.
- We build a discrete-event simulator of RDMA message delivery. It compares go-back-N, selective repeat, selective repeat with a tail-loss probe, and FEC with measured decode costs, and it reproduces 17 hand-derived completion times exactly (§3).
- We show that the decoder flips the winner. FEC with ISA-L's decoder loses to retransmission at RTTs up to 10 µs; FEC with erasure-count decoding wins at every RTT we test. We also show how much parity each loss rate needs, and that the result holds under bursty loss calibrated to Meta's rack measurements [3].

We do not propose a new decoding algorithm. Erasure-count decoding of systematic Cauchy codes is known [7, 10, 12, 2] and implemented in cm256 [11]. Our contribution is to measure what its absence costs in a widely deployed library, and to show when that cost decides the transport design.

2 Background and Related Work

Decoding systematic Reed–Solomon codes. A systematic RS(n,k) code sends k data packets followed by $m = n-k$ parity packets. Any k received packets suffice. The standard decoder, used by ISA-L's documented decode path, gathers the rows of the generator matrix for the first k survivors, inverts this $k \times k$ matrix, and multiplies the m' rows for the lost data by the received packets. The inversion costs $O(k^3)$ field operations; the packet work costs $O(m'kL)$ for packet size L. When only $m' \ll k$ data packets are lost, the received generator has the block form $\left[\begin{smallmatrix} I & 0 \\ A & B \end{smallmatrix}\right]$, and only the $m' \times m'$ Cauchy block B needs inverting [7, 10]. Li [7] inverts B in closed form in $O(m'^2)$, cm256 [11] uses an $O(m'^2)$ LDU decomposition, and the related literature traces the idea further back [12, 2]. Packet work in all of these runs on split-nibble SIMD table lookups [9], or on GFNI instructions in recent ISA-L.

FEC versus retransmission. SDR-RDMA [6] shows that erasure coding beats selective repeat for RDMA over long-haul

links (25 ms RTT). It benchmarks its MDS encoder against ISA-L, assumes encoding overlaps transmission, and does not quantify decoding latency. LoWAR [14] applies FEC to RDMA over lossy WANs. Within the datacenter, Cloudburst [13] uses LT codes over multiple paths to cut TCP tail latency relative to DCTCP. None of these measures decode latency against microsecond RTTs, and none compares against selective repeat with a tail-loss probe [1], the cheapest fix for the tail losses that FEC targets.

3 Method

Decode benchmark. All decoders decode the same codewords: ISA-L is given the same Cauchy generator, and every decode is checked byte-for-byte against the original data. cm256 uses its own generator, but sees the same data and loss patterns. Before each timed phase the packets are placed in a defined cache state, warm by default, which models a NIC writing into the last-level cache. This matters: without it, a decoder with a slow matrix phase sees its data evicted while it inverts, and is charged extra packet time unrelated to its kernel. We report the median of 10,000 decodes on one pinned core, and split ISA-L's time into matrix inversion and packet work. The packet work alone equals the cost of ISA-L with a pre-cached decode matrix.

Simulator. The simulator models one message, or 64 parallel messages forming one collective step, over a 100 Gbps link with 4 KB packets. One-way delay is RTT/2. The sender transmits back-to-back at line rate, and acknowledgements are never lost. It compares four schemes:

- **GBN**: the receiver drops out-of-order packets and NAKs each gap once.
- **SR**: IRN-style [8]; the receiver keeps out-of-order packets and NAKs each hole once.
- **SR+TLP**: SR plus one tail-loss probe 2×RTT after the last new packet.
- **FEC**: systematic RS(n,k), completing a codeword when its data arrives or when any k packets arrive and the measured decode time has elapsed on one decoder core. A codeword that loses more than m packets requests its missing data once its last packet arrives.

The timeout is 100 µs for every scheme. Losses are Bernoulli, or follow a two-state burst model that advances in wall-clock time, so a burst can end while a sender waits for a timeout. All schemes see the same random loss draws. We validated the simulator against 17 hand-derived completion times, covering mid-message and tail losses, repeated losses, fallback and probes; it matches all of them to within 2 ns.

4 Evaluation

4.1 Setup

Decode cost. We measure single-core median decode latency for 4 KB packets on an AWS c7i.2xlarge instance (Intel Xeon Platinum 8488C, Sapphire Rapids, AVX-512 and GFNI, Ubuntu 24.04), with Intel ISA-L v2.31 and cm256. We repeat every measurement on an Intel Core i9-14900HX laptop as a cross-check. All decoders receive identical data, loss patterns and survivor sets, and every decode is verified byte-for-byte. We compare three decoders:

- ISA-L's documented decode path, which inverts the $k \times k$ survivor matrix for every loss pattern;
- cm256, which eliminates the received data and solves an $m' \times m'$ Cauchy system by LDU decomposition;
- erasure-count decoding [7] on ISA-L's GFNI kernel.

Network model. We use the simulator of §3 with Bernoulli loss unless stated otherwise. FEC decode times are the measured server medians, queued on one core. For each point we report the best of the codes RS(17,16), RS(18,16), RS(20,16), RS(34,32) and RS(36,32), and the 99th-percentile completion time over 5,000 messages (1,000 collective steps).

4.2 Decode cost

On the server, ISA-L takes about 21 µs for a cold decode at k=16 and about 160 µs at k=32, almost all of it in matrix inversion; its packet work takes 0.8–3.9 µs. Erasure-count decoding on the same kernel removes almost all of the inversion cost. With one or two lost packets and $k \geq 16$ it decodes 17–91× faster than ISA-L, and between 23% slower and 15% faster than ISA-L with a pre-cached decode matrix. Against cm256 it is 1.0–1.5× faster when one packet is lost ($k \geq 16$) and 2.1–2.6× faster when two are lost. That gain comes from the GFNI kernel, not the decoding algorithm, since both decoders solve the same $m' \times m'$ system. The laptop shows the same pattern with a smaller gap: ISA-L's scalar inversion is about 2–3× faster there, so erasure-count decoding is 11–43× faster than ISA-L instead of 17–91×.

4.3 When does FEC beat retransmission?

Decode cost decides the winner. Table 1 shows a 64-flow collective step at 10^{-3} loss. With ISA-L's $k \times k$ decoder, FEC *loses* to selective repeat with a tail-loss probe at every RTT up to 10 µs, by 1.5–3.0× (48.0 vs. 15.8 µs at RTT 2 µs). The same FEC scheme with an erasure-count decoder *wins* at every RTT, by 1.2–3.0×, and comes within 5–8% of an FEC scheme with free decoding. Neither the code nor the loss process changes between these runs, only the decoder. The same reversal appears at lower loss: at 10^{-4} and RTT 5 µs, erasure-count FEC takes 14.5 µs, retransmission 18.3 µs, and FEC with ISA-L 34.4 µs. With the laptop's faster ISA-L inversion the ISA-L column

Table 1: p99 completion of a collective step (slowest of 64 flows, 128 KB each) at loss rate 10^{-3}, in µs, with decode costs measured on the server. "Best retransmission" is the better of SR and SR+TLP.

RTT (µs)	Best retransmission	FEC, ISA-L	FEC, cm256	FEC, erasure-count	FEC, free decode
2	15.8	48.0	13.1	13.0	12.1
5	23.3	49.5	14.6	14.5	13.6
10	35.8	52.0	17.5	17.3	16.1
25	73.3	59.5	25.0	24.8	23.6

drops to 27.6–39.5 µs at 10^{-3} and FEC with ISA-L still loses at RTTs of 2 and 5 µs.

Why retransmission loses the tail. A lost final packet produces no out-of-order arrival to trigger a NAK, so plain SR and GBN wait for the timeout. Their p99 for a 64-flow step jumps from 18 to 118 µs between 10^{-4} and 10^{-3} loss. A tail-loss probe recovers most of this, but not when the probe or several final packets are lost. At 10^{-2} loss SR+TLP again reaches 118 µs, while erasure-count FEC stays at 16.7 µs and FEC with ISA-L at 50.3 µs.

When FEC does not pay. Parity costs bandwidth: 6.25% for RS(17,16) and RS(34,32). Below about 10^{-4} loss, or for single messages of 2 MB and up, this overhead outweighs the recovery benefit. For example, a single 2 MB message at 10^{-3} loss and RTT 5 µs completes in 173 µs with SR and in 181 µs with FEC. Figure 2 maps the boundary. FEC with an erasure-count decoder wins for small messages at loss rates of 10^{-3} and above, and the region where it wins grows with RTT. ISA-L decoding shrinks that region to almost nothing at RTTs of 5 µs or less.

How much parity. Figure 3 plots tail latency against parity overhead. At 10^{-4}–10^{-3} loss the 6.25% codes are best: RS(17,16) completes a 64-flow step of 128 KB in 14.5 µs, against 23.3 µs for the best retransmission. At 10^{-2} loss they no longer carry enough parity. Too many codewords lose more than one packet, fall back to retransmission and hit the timeout (118.6 µs). RS(20,16), with 25% overhead, completes in 16.7 µs. The right amount of parity therefore grows with the loss rate. For single messages of 2 MB no amount of parity beats retransmission below 1% loss.

Bursty loss calibrated to production measurements. Losses in datacenters are bursty. Ghabashneh et al. [3] measured Meta racks:

- bursts last a median of 2 ms (90th percentile 8 ms);
- they occur 7.5–39.8 times per second;
- 0.4–5% of them drop packets.

We convert these numbers into a two-state burst model that advances in wall-clock time, with lossy bursts of mean 2.9 ms occupying 2.2×10^{-4} (typical rack), 1.15×10^{-3} (busy) or 5.8×10^{-3} (contended) of the time. The loss rate inside a burst is not published, so we sweep it. The same reversal appears (Figure 4). For a 64-flow step of 128 KB in a busy rack with 10% loss inside bursts, the best retransmission takes 25.0 µs, FEC with ISA-L 51.7 µs, and FEC with erasure-count decoding 17.1 µs. In a contended rack with 1% in-burst loss the figures are 18.3, 34.4 and 14.5 µs. At 30% in-burst loss every scheme degrades, because no code we evaluate carries enough parity for a multi-millisecond burst.

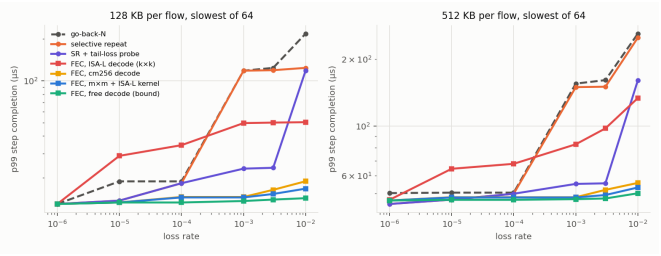


Figure 1: p99 completion of a 64-flow collective step versus loss rate (RTT 5 µs, 100 Gbps).

Which erasure-count decoder matters little. cm256 and erasure-count decoding on ISA-L's kernel differ by at most 2.2 µs at the 99th percentile in the collective configurations above (3.7 µs across the full laptop sweep), even though cm256 decodes up to 2.6× slower. Once decoding costs about 1 µs, RTT and serialisation dominate. The decisive difference is $k \times k$ inversion versus erasure-count decoding, not which erasure-count implementation is used.

4.4 Limitations

Our model has no congestion control, PFC or NIC pipeline. The flows in a collective step are independent and do not share a bottleneck, the sender is not window-limited, and acknowledgements are never lost. Decode costs are single-core medians from one cloud server and one laptop, without frequency pinning (EC2 guests do not expose it). In the bursty model, bursts are independent across flows, whereas a real incast burst hits every flow to the same receiver at once, and the in-burst loss rate is swept rather than measured. Commodity RoCE timeouts are often far above 100 µs, which would favour FEC and SR+TLP further.

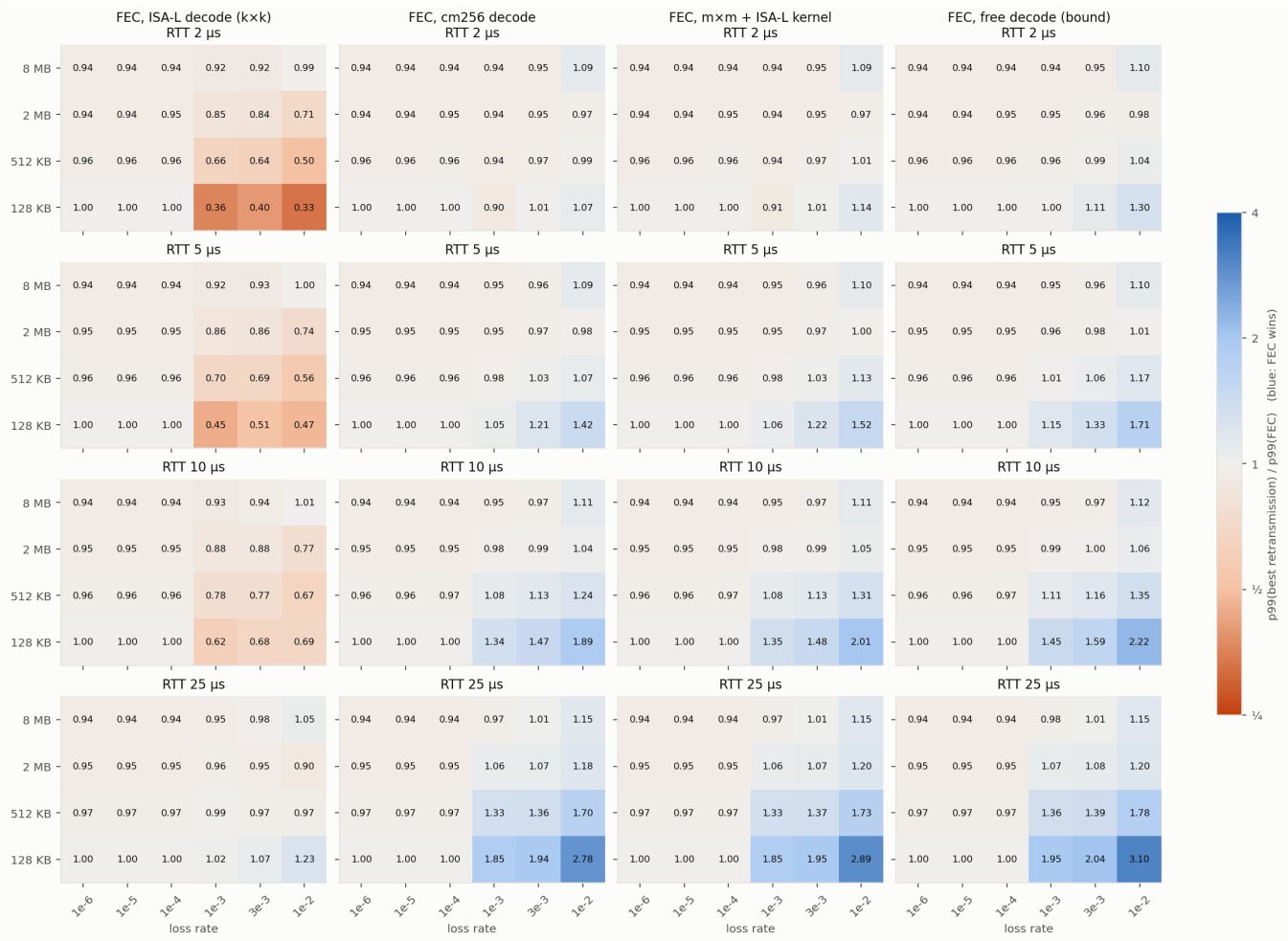


Figure 2: p99(best retransmission) / p99(best FEC code) for a single flow, over loss rate and message size. Rows are RTTs and columns are decoders. Blue means FEC wins.

5 Discussion

What a designer should take away. FEC is worth its bandwidth for small messages and collective steps at loss rates of 10^{-4} and above, but only with a decoder whose cost does not grow with k^3. Off-the-shelf use of ISA-L's documented decode path is enough to turn a 1.6× win into a 2.1× loss at RTT 5 µs. Caching decode matrices recovers most of ISA-L's cost when loss patterns repeat. Erasure-count decoding makes a cache miss cost about the same as a hit, which matters because a code such as RS(36,32) has tens of thousands of distinct loss patterns.

What is missing. Our results come from simulation with measured decode costs, not from an RDMA testbed. Flows in a collective step are independent, so they do not share incast bursts or a bottleneck queue. The in-burst loss rate in our calibrated model is swept, not measured. A NIC-offloaded decoder would shift the boundary again. Hardware validation is the obvious next step.

6 Conclusion

At microsecond RTTs, whether erasure coding beats retransmission for RDMA depends less on the code than on the decoder. With ISA-L's $k \times k$ decode path, FEC loses to selective repeat with a tail-loss probe; with erasure-count decoding, a known but not universally implemented technique, it wins by up to 3× on collective steps and comes within 5–8% of free decoding. The benchmark, simulator and all scripts that produce our figures are available at `https://github.com/1brahim74/thesis`.

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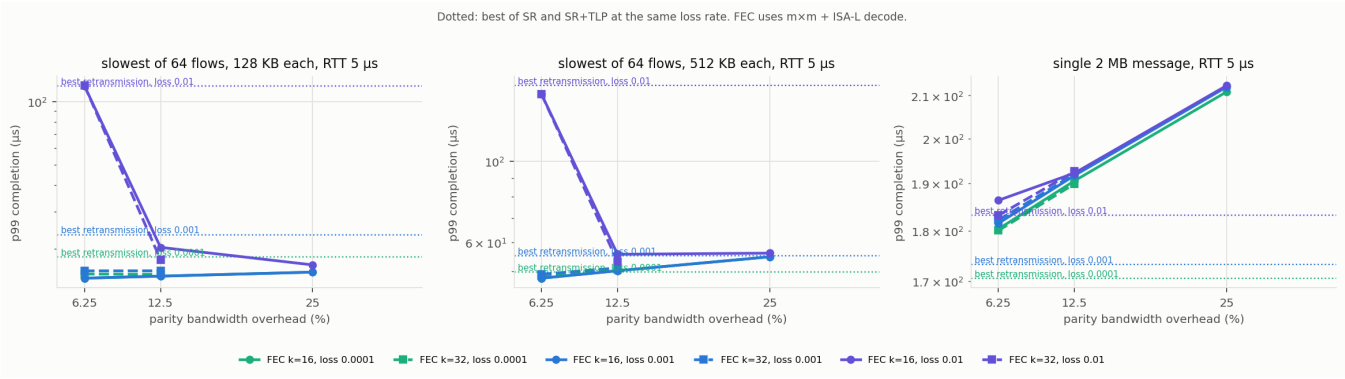


Figure 3: p99 completion versus parity overhead for each code (erasure-count decoding on ISA-L's kernel). Dotted lines: best retransmission at the same loss rate.

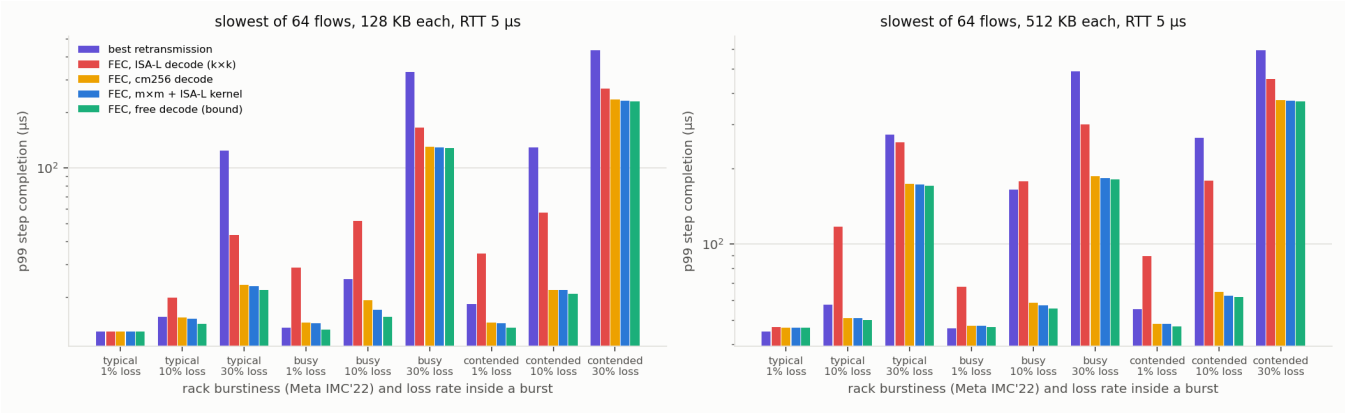


Figure 4: p99 completion of a 64-flow step under bursty loss calibrated to Meta rack measurements [3], for three rack types and three in-burst loss rates.

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